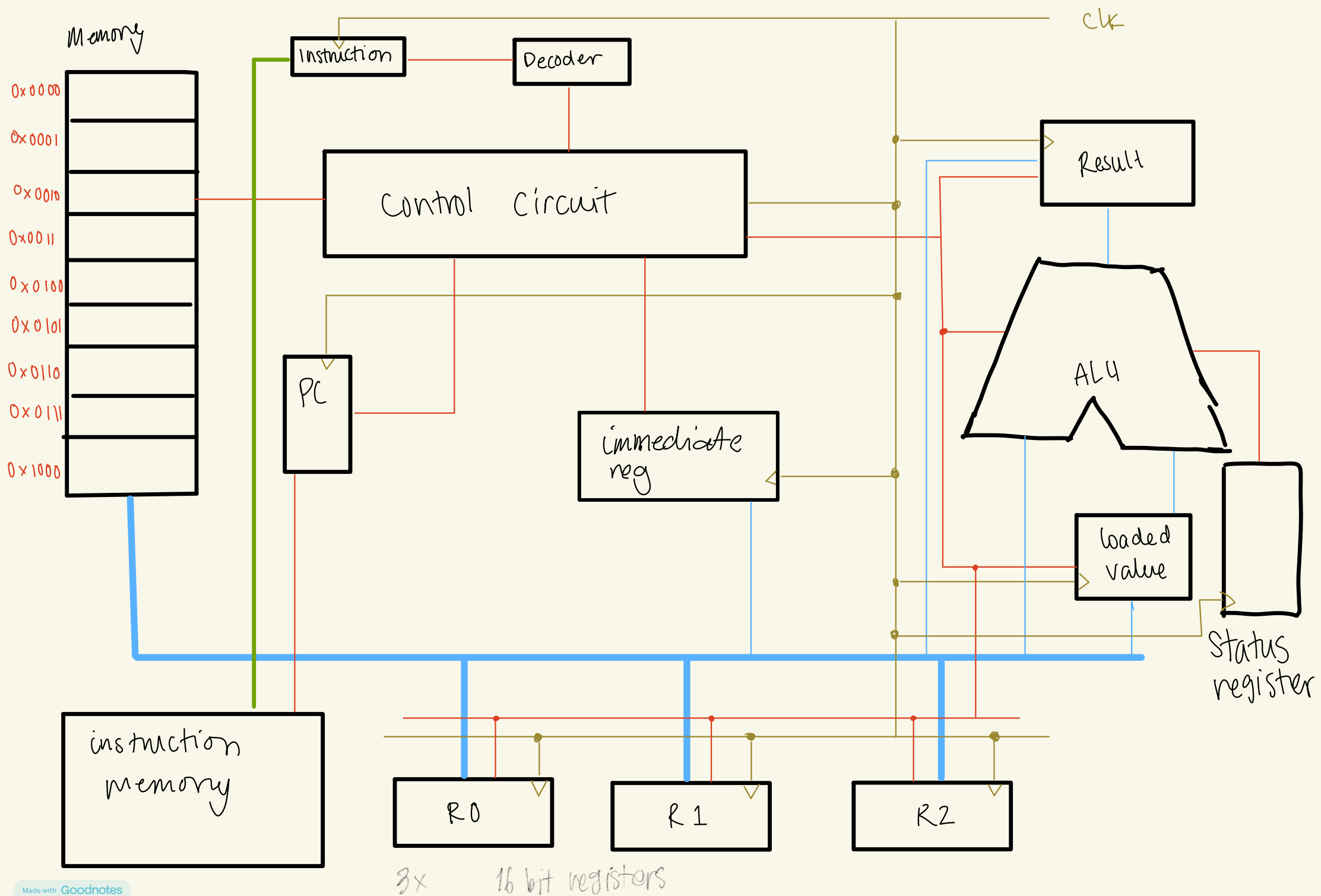


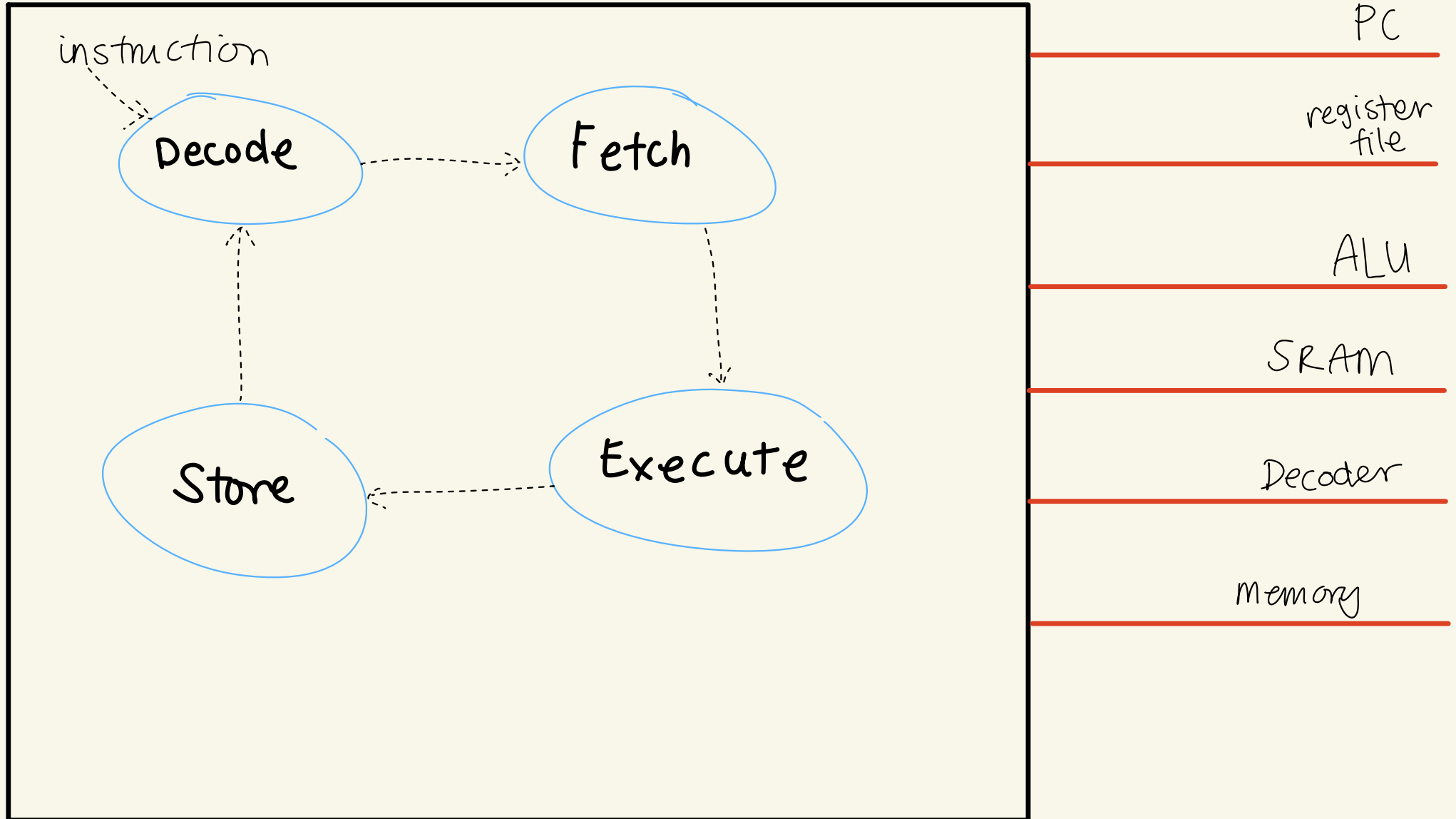


Opcode

INSTRUCTION	PREFIX	SUFFIX	ARGUMENTS
Add	0xA	0x1	R[n:0] x2
Sub	0xA	0x2	R[n:0] x2
Inc	0xA	0x3	R[n:0]
Dec	0xA	0x4	R[n:0]
Lsr	0xA	0x5	R[n:0] & immediate
Lsl	0xA	0x6	R[n:0] & immediate
Xor	0xA	0x7	R[n:0] x2
And	0xA	0x8	R[n:0] x2
Nor	0xA	0x9	R[n:0] x2
Not	0xA	0x10	R[n:0]
Or	0xA	0x11	R[n:0] x2
Ldi	0xD	0x1	R[n:0] & immediate
Ld	0xD	0x2	R[n:0] & mem addr
St	0xD	0x3	R[n:0] & mem addr
Mov	0xD	0x4	R[n:0] x2
Jmp	0xC	0x1	PMEM & SREG
Je	0xC	0x2	PMEM & SREG
Jg	0xC	0x3	PMEM & SREG
JL	0xC	0x4	PMEM & SREG
CP	0xC	0x5	R[n:0] x2

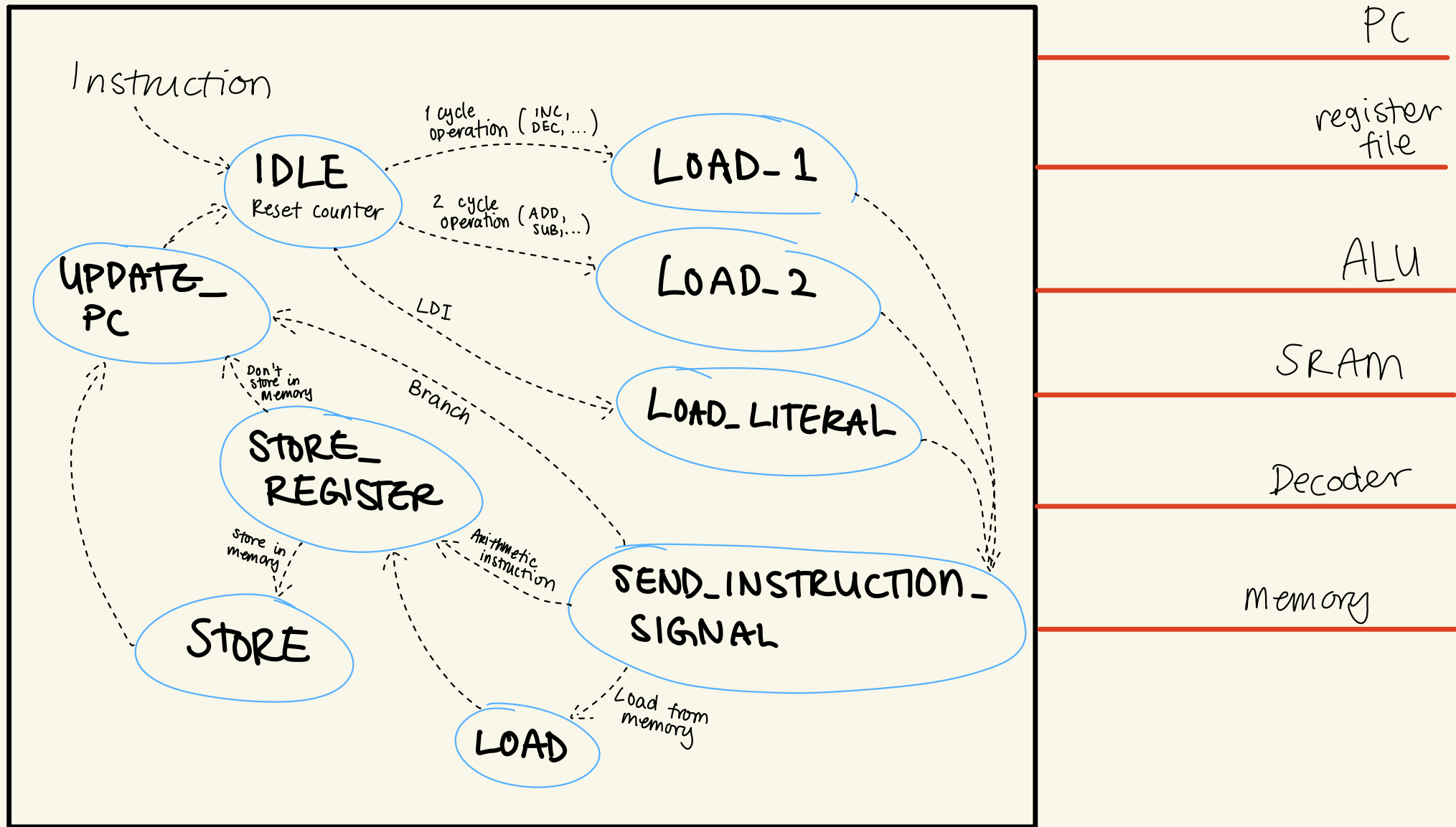
High-level FSM

CONTROL CIRCUIT



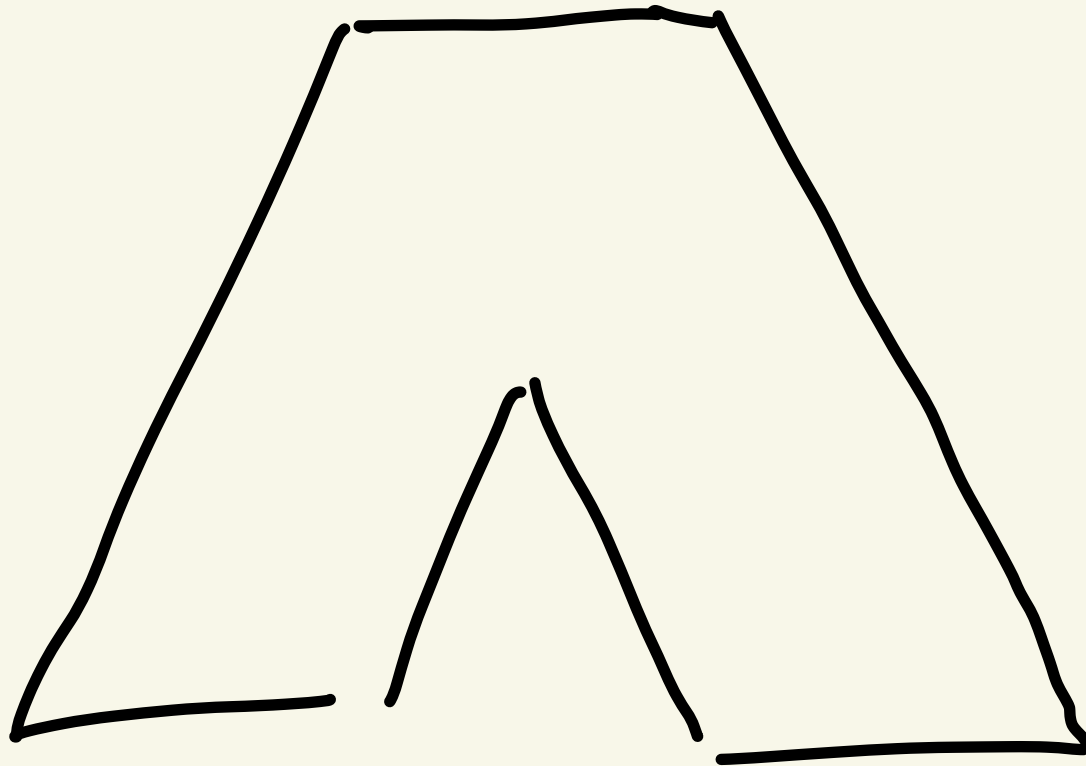
Low-Level FSM

CONTROL CIRCUIT



ALU

Supported instructions



- Adder

↳ add

↳ 2's complement

1. flip bits

2. add 1

2 clock cycles

- Shift register (16 D flip-flops)

↳ LSL

↳ LSR

} ask Rudy!

- Bitwise operators

↳ OR, AND, XOR, NOT

↳ simply a truth table

- Comparison

↳ Status register

↑
again ask Rudy